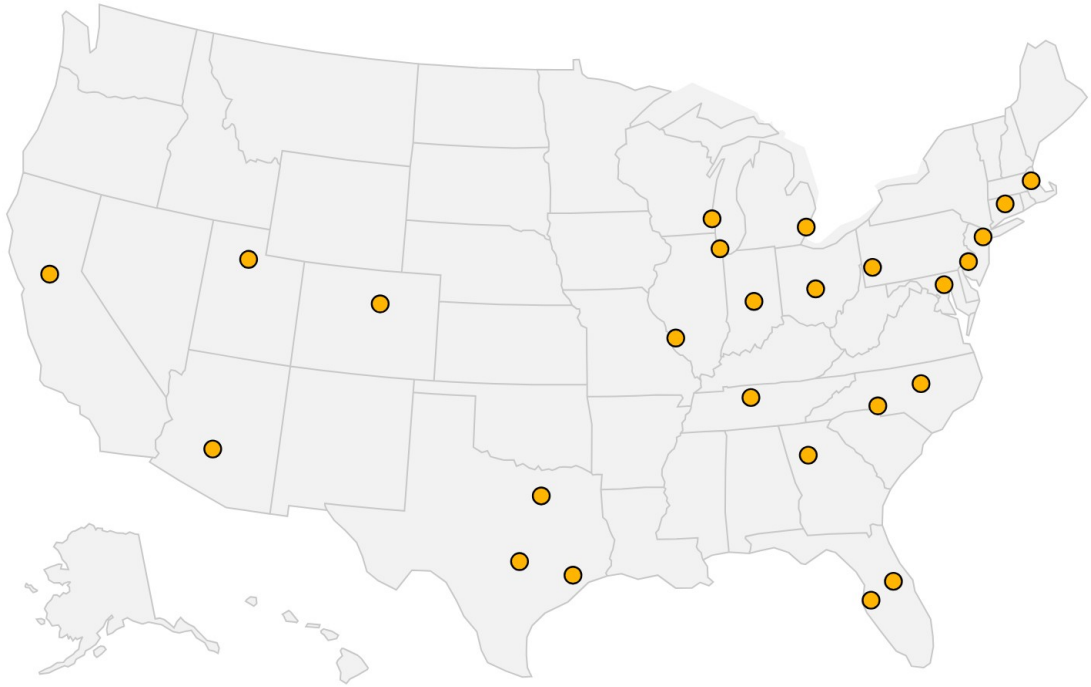
Stockflow

Branch-level demand forecasting and inventory rebalancing

Right product at the right branch, so surgeons, hospitals and their patients are not left waiting.

Bryce Biyeba
Data Science Intern · October 2026



25 branches · 300 SKUs · 4,889 branch × SKU combinations

62% of inventory dollars sit above target while 419 branch positions face stock-outs

5.8%

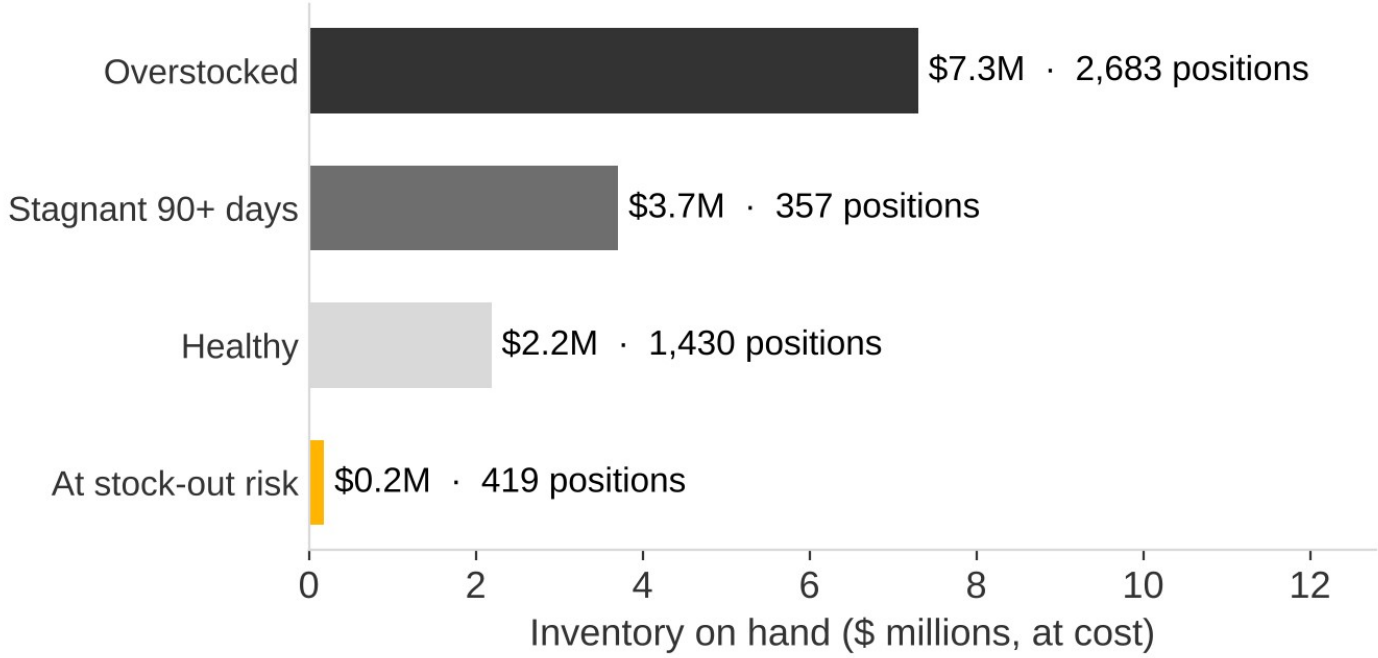
of branch × SKU days were stock-out days (3 years)

$8.3M

above target, of $13.4M on hand

$3.7M

stagnant: no sale in 90+ days



Inventory on hand by position status, at cost

One pipeline turns three years of branch data into transfers a planner approves



sales, stock moves, POs, promotions

outages, returns, stock-out days

LightGBM + statistical, P10 / P50 / P90

safety stock, reorder point, risk

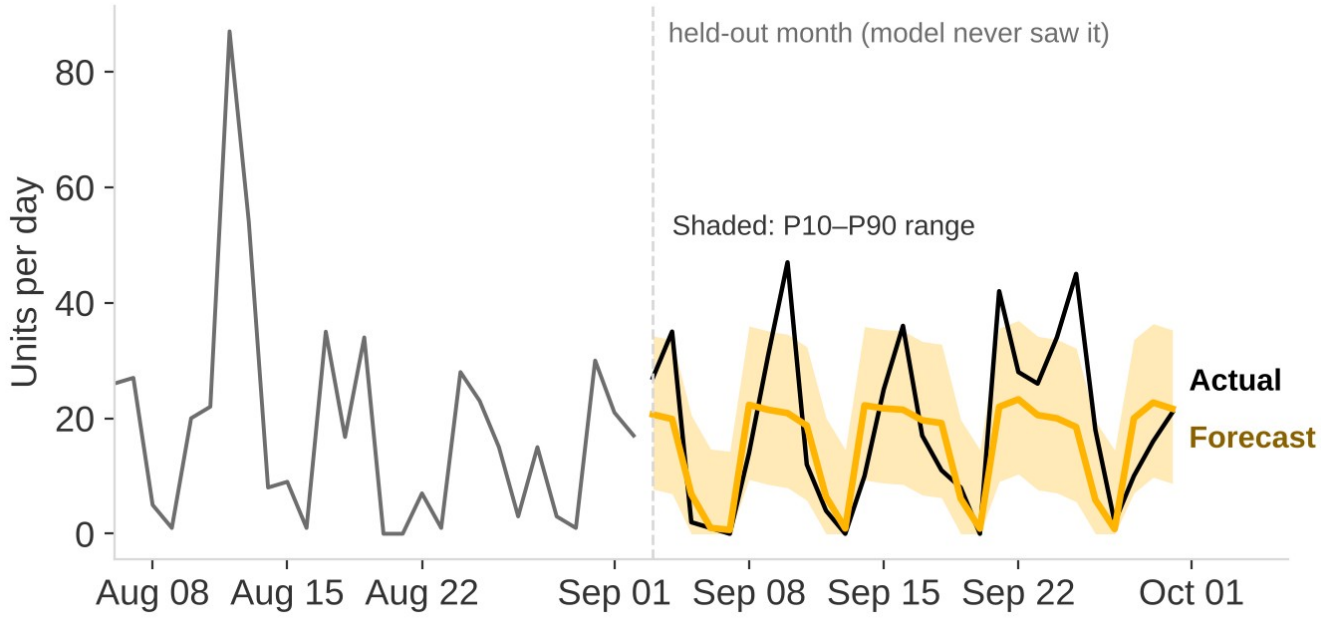
min-cost transfers under constraints

plain-language reason per move

nothing moves without sign-off

Every step checks its inputs; every assumption is documented in the repository.

Forecasts are 33% more accurate than the seasonal baseline on a month the model never saw

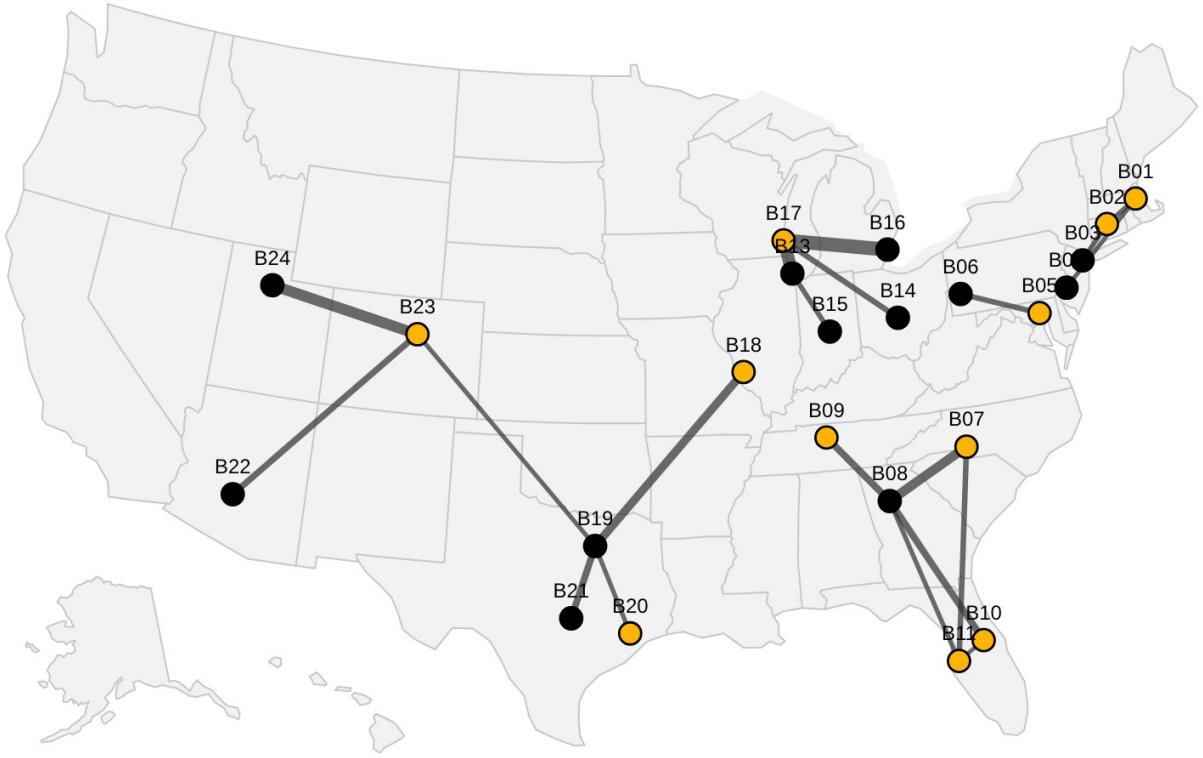


Branch B24, SKU S0057 (Tools), 2026-09-03 to 2026-09-30

Model (held-out month)	WAPE	MAE	Bias
Seasonal naive (baseline)	74.9%	3.38	+1.2%
Exponential smoothing	56.6%	2.55	+2.5%
Croston (intermittent)	57.3%	2.58	+0.9%
LightGBM (global ML)	49.8%	2.25	+0.0%
Selected per segment	**49.8%**	**2.25**	**+0.0%**

WAPE = absolute error ÷ actual units; MAE in units per day. P10–P90 band held 83% of actual days (target 80%).

895 transfers cost $46k to move and save $216k net, $11k more than a rule of thumb



Top 20 of 259 corridors · line width = units · black = sending branch, gold = receiving

From → To	SKU	Units	Net savings	Reason
B10 → B17	S0190	208	$4,318	Excess → shortfall
B19 → B23	S0190	215	$4,206	Excess → shortfall
B14 → B19	S0285	202	$3,781	Excess → shortfall
B10 → B07	S0190	132	$3,330	Excess → shortfall
B19 → B09	S0134	216	$3,249	Excess → shortfall

Each move ships with a plain-language reason a branch manager can check.

Rebalancing frees $1.2M of excess and avoids 85% of expected lost sales

$1.2M

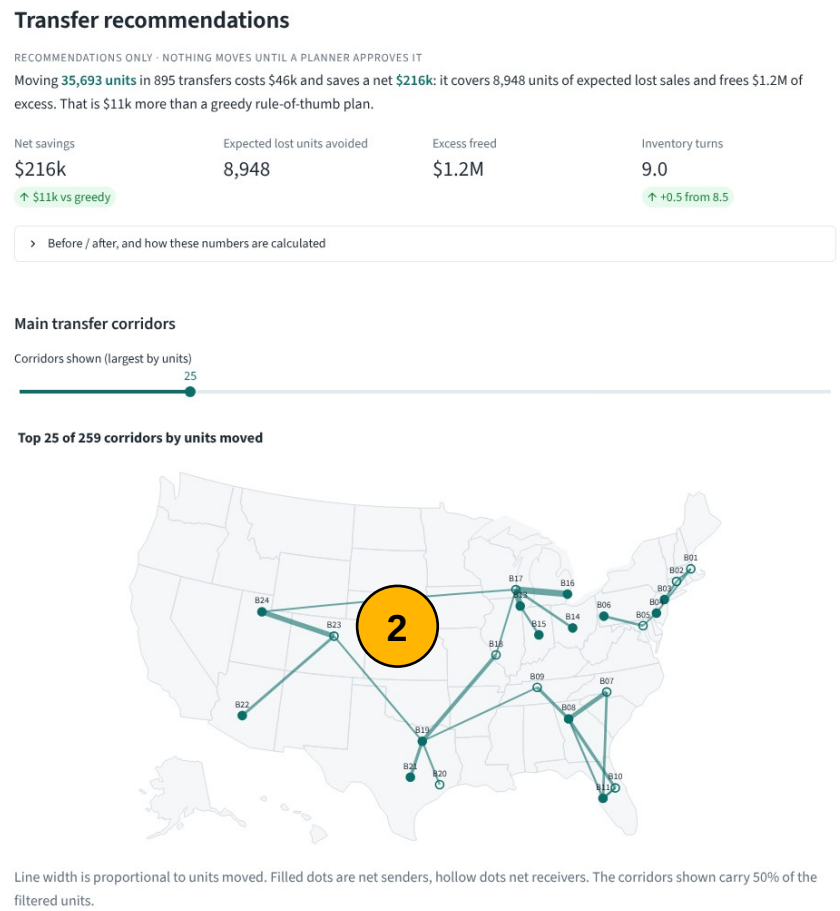
excess inventory put back to work

8,948 lost-sale units avoided

Measure	Before	After
Excess above target	$8.3M	$7.1M
Annual carrying cost of excess	$2.18M	$1.90M
Expected lost-sale units (lead time)	10,533	1,585
Positions at stock-out risk	419	76
Inventory turns	8.5	9.0
Net savings (one cycle)	–	$216k

Model estimates for one planning cycle versus doing nothing.

A branch manager sees what to move, why, and approves it on one screen



3

From	To	SKU	Decide	Note	Status	Reason
B10	B17	S0190			Pending	Excess to shortfall
B19	B23	S0190			Pending	Excess to shortfall
B14	B19	S0285			Pending	Excess to shortfall
B10	B07	S0190			Pending	Excess to shortfall
B19	B09	S0134			Pending	Excess to shortfall
B16	B17	S0134			Pending	Excess to shortfall
B18	B15	S0283			Pending	Excess to shortfall
B09	B19	S0285			Pending	Excess to shortfall

1. The plan in one sentence
2. Where stock flows
3. Approve or reject; logged

Live demo (view-only): stockflow-six-mu.vercel.app

Four challenges shaped the design, and each one left the tool stronger

Stock-out days hide real demand

Empty shelves on 5.8% of branch-product days record zero sales, not zero demand.
Fix: flagged and excluded those days from training and scoring.

The forecast ran 2.6% low at 10× scale

My first theory was wrong; an experiment ruled it out before I built anything.
Fix: per-segment correction from past months only. Bias −2.6% → −0.8%.

The optimization grows fast with scale

Every product can move between every pair of branches.
Fix: pruned routes that can't pay off; parallel product batches. 10×: 160 batches in ~15 min.

Planners need moves they can check

Accuracy alone doesn't get a transfer approved.
Fix: a plain-language reason per move, human approval, and a sensitivity check.

What I learned: test the idea first, and trust is built by showing your work

Test the hypothesis before the fix

My first theory for the bias was wrong. A quick experiment saved days of fixing the wrong thing.

Beat the baseline; say where you don't

The model still loses on 19% of series, mostly slow movers. Showing it made the rest more credible.

Explainability drives adoption

A reason a branch manager can check is worth more than one more point of accuracy.

Delete work that doesn't earn its place

A branch-level correction added noise without helping, so I removed it.

With more time and resources: real data, longer horizons, and closing the loop

Real data and agreed costs

Run on real ERP extracts, and agree the cost of a lost sale and freight rates with finance.

Plan weeks ahead, not one cycle

Optimize transfers and purchase orders together over several weeks.

Better uncertainty

Use the model's own forecast ranges and real lead-time variability for safety stock, not a normal error model.

Close the loop

Raise approved transfers in the ERP, retrain on a schedule, and learn from planner rejections.

Next: a 12-week pilot at 2–3 branches on real ERP and warehouse data

Weeks 1–4

Connect data

7 standard extracts, validated on load

Weeks 5–10

Shadow pilot

2–3 branches; planners approve or reject

Weeks 11–12

Decide

Monitor error and bias; retrain on drift

Limitations

- Synthetic data so far
- Plans one replenishment cycle
- Normal error model for safety stock
- Costs need finance approval

Right product, right branch: surgeons don't wait, so patients don't either.